Two multi-affine equations with exponentially many real connected components

Principia Math In Progress

22 September 2026

MathDB 405042. Negative answer to the degree-only bound.

Task 1: the original question and its answer

The question in Basu's contribution to Oberwolfach Report 9/2025 [1, pp. 445–446] asks whether two real multi-affine equations of bounded total degree have a number of real connected components bounded independently of the number of variables. Here multi-affine means degree at most one in each individual variable. The answer is no, already for total degrees two and four.

Theorem 1. *For every integer $m \geq 2$, there are two multi-affine polynomials of total degrees two and four in $2m$ real variables whose common real zero set has exactly 2^m connected components. Every component is homeomorphic to $\mathbb{R}^m$.*

Proof. Use independent variables $x_1, y_1, \ldots, x_m, y_m$, and put

$$z_i = x_i y_i, \qquad F_m = \sum_{i=1}^{m} z_i - m, \qquad G_m = \sum_{1 \leq i < j \leq m} z_i z_j - \binom{m}{2}. \tag{1}$$

Every nonconstant monomial of F_m is $x_i y_i$. Every nonconstant monomial of G_m is $x_i y_i x_j y_j$, where $i \neq j$. Thus no variable occurs with exponent greater than one in either polynomial. Their total degrees are exactly two and four.

The polynomial identity

$$\sum_{i=1}^{m} (z_i - 1)^2 = F_m^2 + 2(m-1)F_m - 2G_m \tag{2}$$

follows by expanding the square of $\sum_i z_i$. At a common real zero of F_m, G_m, its left side vanishes, so $z_i = 1$ for every i. Conversely, these equalities imply $F_m = G_m = 0$. Consequently the common real zero set is exactly

$$V_m = \{(x, y) : x_i y_i = 1 \text{ for every } i\} \cong (\mathbb{R} \setminus \{0\})^m. \tag{3}$$

The homeomorphism sends $(x_1, \ldots, x_m)$ to $(x_1, x_1^{-1}, \ldots, x_m, x_m^{-1})$; its inverse is the projection to the x-coordinates.

Each x_i is nonzero on V_m. The map taking a point to the sign vector of its x-coordinates is continuous into the discrete set $\{-1, 1\}^m$. Hence a connected subset has a constant sign vector. For each $\epsilon \in \{-1, 1\}^m$, the corresponding fiber is homeomorphic to $\mathbb{R}^m$ by

$$t \longmapsto (\epsilon_1 e^{t_1}, \epsilon_1 e^{-t_1}, \ldots, \epsilon_m e^{t_m}, \epsilon_m e^{-t_m}). \tag{4}$$

Its inverse is $t_i = \log|x_i|$. Thus all 2^m fibers are nonempty and connected, and they are precisely the connected components. □

The degrees stay bounded by four while m is arbitrary, so (3) rules out any finite bound depending only on those degrees. Only the subsequence of even ambient dimensions is needed. An unused free variable gives the same component count in dimension $2m+1$. The coefficients are allowed to depend on m, as in the source question. Identity (2) is a proof device: its square is not being introduced as one of the two defining equations.

A smooth strengthening

The conclusion does not depend on the singularity of the pair (1). Fix $0 < \rho < 1$ and replace its second equation by

$$G_{m,\rho} = G_m + \rho/2 = 0. \tag{5}$$

Corollary 1. *For every $m \geq 3$, the two equations $F_m = G_{m,\rho} = 0$ define a smooth real submanifold of dimension $2m-2$, with exactly 2^m connected components. Each component is diffeomorphic to $\mathbb{R}^m \times S^{m-2}$.*

Proof. Identity (2) gives an exact description in the z-variables:

$$Z_{m,\rho} = \left\{ z : \sum_i z_i = m,\ \sum_i (z_i - 1)^2 = \rho \right\}. \tag{6}$$

This is a sphere of radius $\sqrt{\rho}$, centered at $(1,\dots,1)$, in the $(m-1)$-dimensional affine hyperplane $\sum_i z_i = m$. In particular it is diffeomorphic to S^{m-2}. It is nonempty, and $z_i \geq 1 - \sqrt{\rho} > 0$ throughout.

Fix a sign vector ϵ. The map from $\mathbb{R}^m \times Z_{m,\rho}$ given by

$$x_i = \epsilon_i e^{t_i}, \qquad y_i = \epsilon_i z_i e^{-t_i} \tag{7}$$

is a diffeomorphism onto the part with these signs. Its inverse is $t_i = \log|x_i|$, $z_i = x_i y_i$. Since $m \geq 3$, the sphere in (6) is connected. Each sign fiber is therefore connected; the continuous sign map again separates the 2^m fibers.

For completeness, the two defining equations are transverse at every real solution. The map $(x, y) \mapsto z$ has surjective differential because $x_i \neq 0$, and its derivatives with respect to the y_i have nonzero diagonal entries x_i. In z-coordinates the two gradients are

$$(1,\dots,1), \qquad \left(\sum_j z_j - z_1, \dots, \sum_j z_j - z_m \right).$$

They can be dependent only if all z_i coincide. On the hyperplane in (6), that would give $z_i = 1$, contradicting $\rho > 0$. Their pullbacks are independent, proving the claimed smoothness and dimension. □

Tasks 2–5: four proposed mathematical follow-ups

Task 2: Determine the least total-degree bound d for which two multi-affine equations can have unboundedly many connected components as the ambient dimension grows; in particular resolve the cases $d = 2$ and $d = 3$.

Task 3: Determine the largest possible number of real connected components of two multi-affine equations of degrees at most four in n variables, or determine its exponential growth rate.

Task 4: Decide whether a degree-only component bound holds when each of the two defining multi-affine polynomials is invariant under every permutation of the n variables.

Task 5: Determine whether two multi-affine equations of uniformly bounded total degree can have a compact smooth common real zero set with an unbounded number of connected components.

These are proposed directions, not additional claimed solutions.

Verification, sources, and scope of review

The accompanying exact verifier checks the identity, monomial degrees, representatives of every sign component in finite dimensions, and rational transverse examples of (5). These checks supplement the proofs for arbitrary m; they do not prove the infinite family. Review is by the authoring agent only, without independent peer review or a proof assistant.

[1] *MFO–RIMS Tandem Workshop: Optimization, Theoretical Computer Science and Algebraic Geometry: Convexity and Beyond*, Oberwolfach Report 9/2025, doi:10.4171/OWR/2025/9. The complete 46-page published report was read. The contribution on printed pp. 445–446 states the unrestricted two-equation question and cites the earlier three-equation construction.

[2] S. Basu and D. Perrucci, *Topology of real multi-affine hypersurfaces and a homological stability property*, arXiv:2204.01595v1 (4 April 2022), all 26 pages read. Its Example 2.2 uses three equations; its two-equation question is open on pp. 6 and 24. The related 33-page journal version, *Adv. Math.* 420 (2023), 108982, was identified but not obtained. The later published source [1] explicitly cites that journal paper and still poses the present question.

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